

Question number	Answer	Additional guidance	Mark
7(a)	• correct magnification (1) • magnification given in correct standard form (1)	$(98 \pm 1 \div 0.00075) = 130\,600$ $1.3 \times 10^5 / 1.31 \times 10^5 / 1.306 \times 10^5$ ACCEPT $97 \div 0.00075 = 1.29 \times 10^5$ $99 \div 0.00075 = 1.32 \times 10^5$ Correct answer scores full marks	(2)

Question number	Answer	Additional guidance	Mark
7(b)(i)	C $3 \text{ nmol mg}^{-1} \text{ min}^{-1}$ A, B and D are not correct because the correct answer is $60 \div 20 = 3$		(1)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	An explanation that includes four of the following points: Sample 1: (carbon dioxide is produced) because pyruvate can be used (as a substrate) for {aerobic respiration / link reaction} (in mitochondria) (1) Sample 2: - ADP increases rate of (aerobic) respiration (1) - because there is (sufficient) ADP for synthesis of ATP (1) Sample 3: - the pyruvate transport inhibitor stops pyruvate being taken into the {mitochondria/matrix} (1)	ACCEPT pyruvate releases carbon dioxide when forming acetyl CoA ACCEPT ADP increases rate of Krebs cycle ACCEPT ADP is converted to ATP ACCEPT ADP is not limiting ACCEPT the pyruvate transport inhibitor stops pyruvate entering {link reaction / Krebs cycle} ACCEPT pyruvate is not converted into acetyl CoA / acetyl CoA doesn't form (in sample 3)	(4)

Question number	Answer	Additional guidance	Mark
7(c)(i)	An explanation that includes the following points: • slow twitch muscle fibres have more mitochondria (than fast twitch muscle fibres) (1) • because they rely on (ATP from) aerobic respiration (1) • (slow twitch fibres have more mitochondria) because oxygen supply is sufficient / more capillaries carrying oxygenated blood / cells contain (more) myoglobin (1)	ACCEPT converse ACCEPT fast twitch relies on (ATP from) {anaerobic respiration / creatine phosphate} ACCEPT converse	(3)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	A description that includes the following points: • ATP binds to the myosin head (1) • {releasing the myosin head from the actin strand/breaking the cross-bridge} (1) • ATP is hydrolysed (by the myosin head) (1) • providing energy to move the myosin head {back/upright} (1)	ACCEPT ATP broken down into ADP ACCEPT change shape of myosin head / return to original position Do not accept the power stroke	(4)